Environment Similarity Index: Quantifying map-environment divergences for robot localization performance estimation and improvement

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***Abstract*—Todo**

***Index Terms*—Localization, Mapping, AMR, AMCL**

I. Introduction

Localization is a critical ability needed by Autonomous Mobile Robots (AMRs) in order to execute missions successfully. The gold standard for indoor industrial localization has for a long time been Adaptive Monte Carlo Localization (AMCL) [ref], typically used with 2D laser scanners. It is multi-modal, robust to noise and outliers, and well suited for non-linear systems. However, it assumes that the environment is static. Dynamic objects that move while being observed will only briefly obscure the static environment behind them, and in most cases AMCL is robust to this. It is also straight-forward to detect and filter out moving objects. Transient objects on the other hand, such as pallets, boxes, bins, etc, pose a bigger challenge. If the majority of the environment does not change, AMCL is very reliable, but when the changes become too great, the particle filter cannot converge, and its performance degrades rapidly [ref].

A natural question to ask is therefore: How well does the environment match the map? What has been added and what has been removed? It is hyopthesized that such a score or index of map-environment simiilarity could be used as a predictor of AMCL's performance in changing environments, and could be used to the performance of AMCL by tuning its parameters dynamically. However, to the best of our knowledge, no such score or index has been proposed or used in the literature.

A. Contribution

This is the main contribution of this paper. We propose the Environment Similarity Index (ESI): An index that quantifies how well the currently observed environment matches the map. We also show how the ESI correlates to changes in the environment, both on simulated and real-world data, and how these changes impacts AMCL's performance. In addition, we show how the ESI can be used by a user to easily identify areas in their site that have changed since mapping, and therefore may increase localiztion errors. Finally, we show how the ESI can be used to improve AMCL's performance by using it to dynamically tune AMCL's parameters. [be more precise here when i know which]

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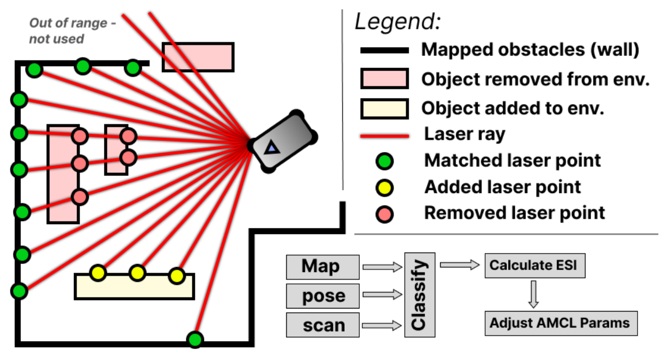

Fig. 1. Example of a figure caption.

II. Related Work

- Bayes filtering / Probabilistic map update - Scan-BIM divergences - Occupancy grid vs feature maps -

III. Methodology

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IV. Experiments

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V. Conclusions

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VI. Limitations and Future work

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A. Bullet list

- Item1
- Item2

B. Equations

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An excellent style manual for science writers is [7].

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TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
copy	More table copy[a]		

[a]Sample of a Table footnote.

Fig. 2. Example of a figure caption.

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Acknowledgment

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

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References

[1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.

[2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[3] I. S. Jacobs and C. P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.

[4] K. Elissa, "Title of paper if known," unpublished.

[5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.

[6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].

[7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.